

Abstract

Background: Premenstrual dysphoric disorder (PMDD) is a DSM-5 depressive disorder affecting an estimated 31 million women and girls worldwide.¹ This scoping review examined psychosocial outcomes and coping mechanisms among women with PMDD and assessed the geographic distribution of the evidence.

Methods: Following Arksey and O'Malley's framework and PRISMA-ScR guidance, we searched PubMed/MEDLINE, CINAHL, APA PsycInfo, APA PsycArticles and the Cochrane Library for peer-reviewed studies published 2010–2025. Eligible empirical and qualitative studies had PMDD as the primary population and at least one psychosocial outcome or coping mechanism as a primary aim.

Results: The search identified 1,943 records; 1,618 remained after deduplication and 44 studies met the inclusion criteria. Depression was most documented (n=39), followed by psychological distress (n=24), interpersonal functioning (n=18), suicidal ideation or self-harm (n=14) and quality of life (n=14). Only three studies examined coping mechanisms. Only three studies (6.8%) originated from low- and lower-middle-income countries, including one from Nigeria.

Conclusions: Evidence on PMDD's psychosocial burden is growing but geographically concentrated in high-income countries, revealing a critical LMIC evidence gap.

Keywords: premenstrual dysphoric disorder; psychosocial outcomes; coping mechanisms; LMICs; scoping review

Introduction, Objective, and Research Question

Introduction: PMDD is a recurring, hormone-linked mood disorder in which irritability, mood swings, depressed mood and anxiety emerge in the late luteal phase and remit after menses begin. It is classified as a depressive disorder in DSM-5² and recognized in ICD-11.³ Using prospectively confirmed diagnoses, community prevalence is about 1.6%, roughly 31 million women and girls worldwide.¹ PMDD has been linked with suicidal ideation and attempts,^{4,5} strained romantic relationships⁶ and reduced quality of life.⁷ Premenstrual symptoms are also common across Africa, where pooled premenstrual syndrome prevalence reaches 47%,⁸ yet it remains unclear how women with PMDD cope or where the evidence has been generated.

Objective: To map psychosocial outcomes and coping mechanisms reported among women with PMDD and identify geographic gaps in the published literature.

Research Question: What psychosocial outcomes and coping mechanisms have been reported in studies of women with PMDD published between 2010 and 2025, and how are these studies distributed across geographic settings?

Methods

Category	Description
Study Design	Scoping review guided by Arksey and O'Malley's framework ⁹ with Levac et al.'s refinements, ¹⁰ reported using PRISMA-ScR. ¹¹
Databases	PubMed/MEDLINE, CINAHL, APA PsycInfo, APA PsycArticles and Cochrane Library.
Publication Years	2010–2025.
Inclusion Criteria	Peer-reviewed empirical or qualitative studies with PMDD as the primary population and at least one psychosocial outcome or coping mechanism as a primary aim.
Exclusion Criteria	Animal studies, laboratory research, reviews, interventions and non-peer-reviewed publications.
Screening Process	1,943 records identified; 325 duplicates removed; 1,618 records screened by title and abstract; 64 assessed for eligibility; 44 studies included.
Evidence Mapping	Included studies categorized by psychosocial outcome, coping mechanism, publication year and geographic setting (country and World Bank income group ¹²).

1,943

records identified
across 5 databases

44

studies included
in the scoping review

3

studies addressing coping
6.8% of 44

3

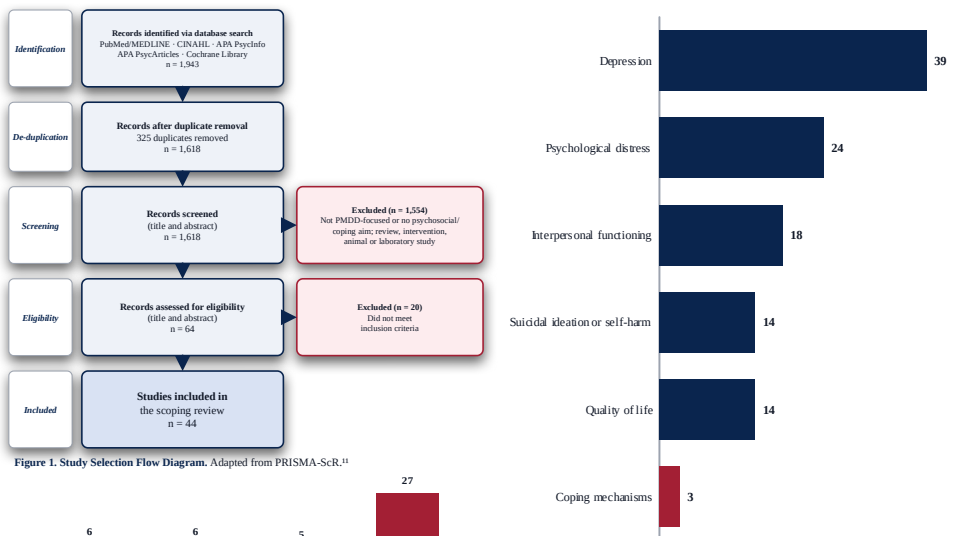
low/lower-middle-income studies
6.8% of 44

Figure 1. Study Selection Flow Diagram. Adapted from PRISMA-ScR.¹¹

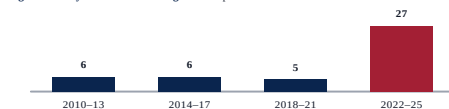


Figure 3. Included Studies by Publication Period. 61% (27 of 44) were published in 2022–2025.

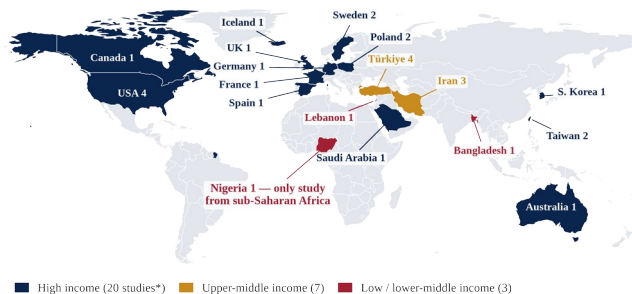


Figure 4. Country Setting and World Bank Income Group of Included Studies. Map shows the 30 studies with an identifiable country setting; income groups follow the World Bank FY2027 classification.¹² The other 14 used online or multinational samples or had no single mappable setting in the available records. *Includes Hong Kong (1).

KEY TAKEAWAY PMDD's psychosocial burden is well documented in high-income countries, yet nearly invisible in low-income settings and sub-Saharan Africa.

Results, Discussion, Conclusion, and Limitations

Results

The database search identified 1,943 records. After 325 duplicates were removed, 1,618 records were screened by title and abstract; 64 were assessed for eligibility, 20 were excluded, and 44 studies were included (Figure 1).

Depression was the most documented outcome (n=39); one nationwide cohort found a 2.6-fold higher risk of later depression after a PMDD diagnosis.¹³ Psychological distress (n=24), interpersonal functioning (n=18), suicidal ideation or self-harm (n=14) and quality of life (n=14) followed. Only three studies examined coping mechanisms, such as peer support shared in online PMDD communities¹⁴ (Figure 2). Most studies (61%) were published in 2022–2025 (Figure 3).

Twenty studies came from high-income countries and seven from upper-middle-income countries (Turkey, Iran); only three (6.8%) came from low- and lower-middle-income countries—Bangladesh,¹⁵ Lebanon¹⁶ and Nigeria,¹⁷ the only study from sub-Saharan Africa (Figure 4).

Discussion
The focus on depression and distress mirrors population studies linking PMDD to depression, suicidal ideation and suicide attempts.^{4–7} Coping, by contrast, appeared in only three studies, so little is known about how women manage symptoms day to day or which supports help.

Evidence is also concentrated in high-income countries. Premenstrual symptoms are common across Africa,⁸ so a single sub-Saharan study likely reflects under-study rather than low burden. Stigma, limited mental health services and local explanations of menstrual distress may shape both burden and coping in these settings.

Patients describe misdiagnosis and uneven provider knowledge,^{18,19} and many turn to online peer communities for support.¹⁴ Because much PMDD research relies on provincial rather than prospectively confirmed diagnosis,¹ future studies should pair confirmed diagnosis with measures of coping and help-seeking.

Conclusion

PMDD carries a well-documented psychosocial burden, especially depression, distress and suicidality, yet how women cope has received little attention, and evidence from low-income settings and Africa is nearly absent. Culturally responsive screening, integrated mental and reproductive health services, and Africa-centered research investment are needed to make this burden visible to health systems.

Limitations

- **Search scope:** studies outside the five databases or the 2010–2025 period may have been missed.
- **Study differences:** variation in design and outcome definitions limits direct comparison.
- **Coding overlap:** a study could address more than one outcome, so counts should not be summed.
- **Geographic coverage:** few studies from low- and lower-middle-income countries limit conclusions about those settings.

Public Health Implications and Future Work

Public Health Implications

- **Screen for suicide risk:** ask about suicidal thoughts when women present with severe premenstrual mood symptoms.^{4,5}
- **Integrate care:** embed PMDD screening in sexual and reproductive, school and university health services.
- **Train providers:** improve recognition to reduce misdiagnosis and delayed care.¹⁸
- **Address coping needs:** ask how women manage symptoms and connect them with peer and professional support.¹⁴

Future Work

- **Confirm PMDD prospectively** using validated daily ratings.^{1,20}
- **Prioritize low-income and African settings**, including qualitative studies of stigma and coping.
- **Validate screening tools** in African languages and test coping-focused interventions.

References



6.8%

of included studies (3 of 44) came from low- and lower-middle-income countries
20 high-income · 7 upper-middle-income
Sub-Saharan Africa: 1 study (Nigeria)